LUKE KURLANDSKI

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EDUCATION

Doctor of Philosophy (Ph.D.) @Rochester Institute of Technology (RIT) *Aug. 2022 - current*
- Department: Computing and Information Science
- Advised by: Dr. Matthew Wright & Dr. Yin Pan

Bachelor of Science (B.S.) @The College of New Jersey (TCNJ) *Aug. 2018 - May 2022*
- Department: Computer Science
- Minors: Mathematics, Physics

SKILLS

Languages:	Python, MatLab, R, Mathematica, Java, C++, C, Rust
Libraries:	PyTorch, huggingface, scikit-learn, TensorFlow
Tools:	Git, SLURM, GNU-Linux, LaTeX
Security:	IDA, redare2, ghidra, OllyDbg

RESEARCH EXPERIENCE

Malware & ML, RIT *Aug. 2022 - current*
Graduate Research Assistant; advised by Dr. Yin Pan & Dr. Matthew Wright.
- Researched malware classification and adversarial malware generation using state-of-the-art deep learning tech.

Learning Curve, TCNJ *Feb. 2021 - May 2022*
Undergraduate Research Assistant; advised by Dr. Michael Bloodgood.
- Researched active learning stopping methods for low-resource natural language processing tasks.

Cybersecurity Division, CACI International *Jun. 2021 - Aug. 2021*
Research Intern
- Researched deepfake audio detection for US Department of Defense (DoD) cybersecurity applications.

Collective Intelligence, TCNJ *Jun. 2020 - Feb. 2021*
Undergraduate Research Assistant; advised by Dr. Monisha Pulimood
- Researched collective intelligence for web-applications and built machine learning recommendation system.

Photonics & Materials Science, TCNJ *Dec. 2018 - Feb. 2020*
Undergraduate Research Assistant; advised by Dr. David McGee
- Researched optical microstructures; wrote APIs for laboratory hardware and GUI applications for experiments.

PUBLICATIONS

- **L. Kurlandski**, H. Berger, Y. Pan & M. Wright, "Beyond Raw Bytes: Towards Large Malware Language Models," *Network and Distributed System Security Symposium (NDSS)*, 2026.
- I. Ryan, **L. Kurlandski** & N. Mathews, "Towards Moving Target Defense for IoT Malware Detection," *Workshop on Software Engineering Research and Practices for the IoT (SERP4IoT)*, 2026.
- **L. Kurlandski**, R. Mosli, Y. Pan, S. Thianphan & M. Wright, "Generating Adversarial Malware by Only Substituting API Calls," *Workshop on Rethinking Malware Analysis (WoRMA)*, 2025.
- **L. Kurlandski** & M. Bloodgood, "Impact of Stop Sets on Stopping Active Learning for Text Classification," *IEEE International Conference on Semantic Computing (ICSC)*, 2022.
- J. Strobelt, D. Stolz, M. Leven, M. Van Soelen, **L. Kurlandski**, H. Abourahma & D. McGee, "Optical Microstructure Fabrication Using Structured Polarized Illumination," *Optics Express*, 2022.
- D. Stolz, J. Strobelt, M. Leven, **L. Kurlandski**, H. Abourahma & D. McGee, "One-Step Fabrication of Surface Relief Dot-Matrix Holograms Using Supramolecular Azopolymer Thin Films," *SPIE Practical Holography: Displays, Materials, and Applications*, 2021.
- J. Strobelt, D. Stolz, M. Leven, **L. Kurlandski** & D. McGee, "Nanoimprint Lithography for the Replication of Optical Microstructures on Azopolymer Thin Films," *IEEE Conference on Lasers and Electro-Optics (CLEO)*, 2021.

POSTERS & TALKS

- **Kurlandski, L.**, Pan, Y. Wright, M. "From Bytes to Insights: A New Era of Malware Detection". In preparation to be presented at *Western New York Image and Signal Processing Workshop (WNYISPW)*. 2023.

- **Kurlandski, L.** "Transferring Benignness to Assembly Instructions". Presented in fulfillment of RIT's research potential assessment for Ph.D. students. 2023.

- **Kurlandski, L.**, Pan, Y. Wright, M. "An Adversarial Malware Evasion Attack Using Machine Translation". Presented at *AI@RIT Summit*. 2022.

- **Kurlandski, L.** "Detecting Deepfake Audio with Deep Neural Networks". Presented in fulfillment of TCNJ's student capstone. 2022.

PROJECTS

MDALh *Jun. 2023 - current*

`github.com/lkurlandski/MDALTH`

- A modular deep active learning library with high-level wrappers for huggingface pretrained models.
- Used Python with PyTorch, transformers, datasets, tokenizers, etc.

Explaining MalConv *Oct. 2022 - current*

`github.com/lkurlandski/MalConv2`

- Identify notable malware byte patterns for MalConv malware classifier using explainable AI techniques.
- Used Python, C++ with PyTorch, captum, pefile, LIEF, capstone, etc.

Adversarial Evasion Attacks *Apr. 2023*

`github.com/lkurlandski/AdversarialEvasion`

- Implemented FGSM and PGD adversarial attacks to evade pretrained image classifiers.
- Used Python with PyTorch, etc.

Website Traffic Fingerprinting *Mar. 2023*

`github.com/lkurlandski/WebsiteFingerprinting`

- Trained deep learning classifiers to perform website traffic fingerprinting.
- Used Python with PyTorch, TensorFlow, etc.

scikit-AL *Jan. 2022 - Aug. 2022*

`github.com/lkurlandski/ActiveLearning`

- A framework for analyzing active learning stopping methods and query algorithms.
- Used Python with scikit-learn, huggingface, ModAL, etc.

Collective Recommender *May 2020 - Nov. 2020*

- A hybrid recommendation system to help users find relevant academic literature.
- Used Python & PostgresSQL with LightFM, NetworkX, pandas, psycopg2, etc.

Tchisla Solver *Jan. 2020 - May 2020*

`https://github.com/lkurlandski/TchislaSolver`
`https://luke-kurlandski.shinyapps.io/TchislaSolver/`

- A dynamic programming algorithm and web application to find solutions for the math puzzle game, *Tchisla*.
- Used Python & R

Newport Hologram Creation *Jan. 2019 - May 2019*

`github.com/lkurlandski/Newport-Hologram-Creation`

- An API to control laboratory equipment and a GUI application to run experiments that create holograms.
- Used Python with PySerial, tkinter, etc.

TEACHING & LEADERSHIP

Instructor, RIT *Jan. 2024 - May 2024*
- Instructed graduate research seminar on Advanced Malware Forensics.

Graduate Teaching Assistant, RIT *Aug. 2023 - Dec. 2024*
- Aid for graduate research seminar on Human Factors in Security and Privacy.

Research Mentor, RIT *Jan. 2023 - current*
- Mentored undergraduate students for capstone projects, summer research experiences, and independent studies.

Lead Undergraduate Research Assistant, TCNJ *Aug. 2021 - May. 2022*
- Student lead among undergraduate RAs (Learning Curve Group).

GED Tutor, TCNJ *Nov. 2017 - May. 2018*
- Taught math/science/english to adults seeking to earn their general equivalency degree.

AWARDS & HONORS

National Science Foundation (NSF) Graduate Research Fellowship Program (GRFP) (recipient) *Apr. 2024*
- Recipient of prestigious national fellowship program.

National Defense Science and Engineering Graduate (NDSEG) Fellowship (recipient) *Mar. 2024*
- Recipient of prestigious national fellowship program. Declined due to conflict with GRFP.

AI Seed Fund (recipient), RIT *Jan. 2024*
- Recipient of local funding competition at RIT.

Goldberg-Neff Scholarship (recipient), TCNJ *May 2022*
- Recipient of scholarship for graduating students pursuing graduate research.

Upsilon Pi Epsilon Computer Science Honor's Society (member), TCNJ *May 2022*
- Inducted as a member into the distinguished national Honor's Society.

Phi Beta Kappa Honor's Society (member), TCNJ *May 2022*
- Inducted as a member into the distinguished national Honor's Society.

Kappa Delta Rho Nu Gamma Chapter (founder), TCNJ *Jan. 2019 - May 2022*
- Established the Nu Gamma Chapter of Kappa Delta Rho fraternity at TCNJ.

Department of Computer Science Award (recipient), TCNJ *May 2019*
- Recipient of department award for first-year students.

Dean's List 8/8 semesters, TCNJ *Aug. 2018 - May 2022*
- Maintained a GPA above 3.5 all eight semesters of undergraduate.